

Visibility Modelling of a Star-Forming Ring in NGC 3351

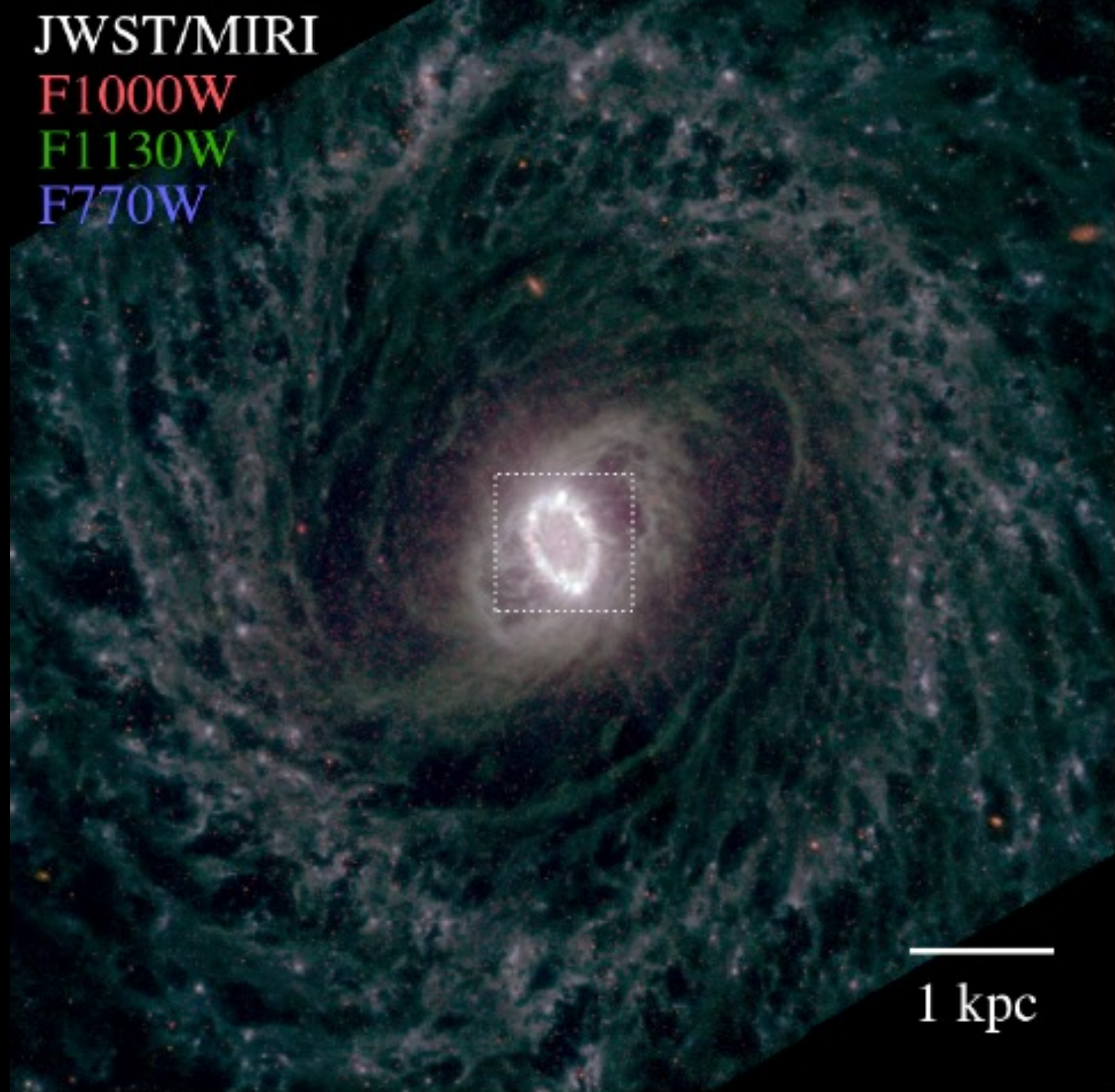
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Lethbridge

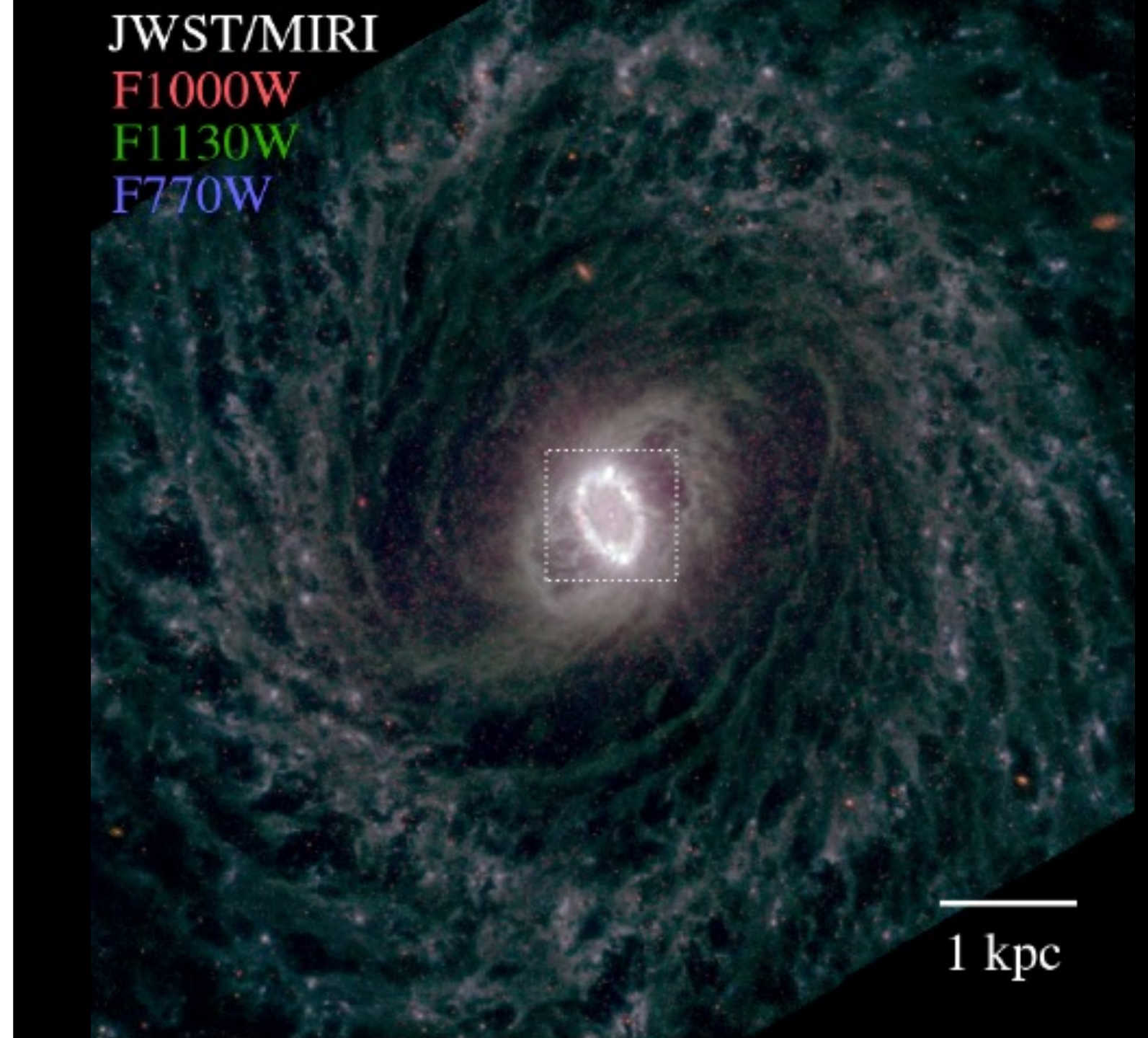
NRC-CMRC

JWST/MIRI
F1000W
F1130W
F770W



1 kpc

What is the
nature of star-
forming regions in
nearby galaxies?

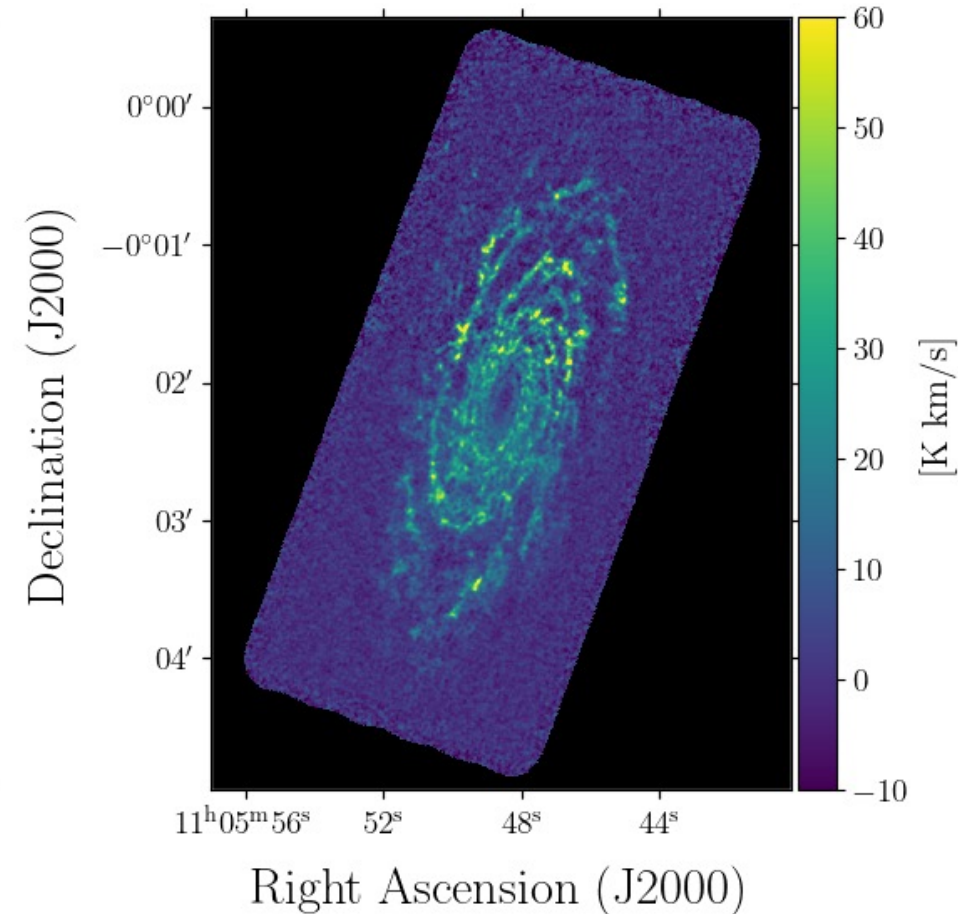
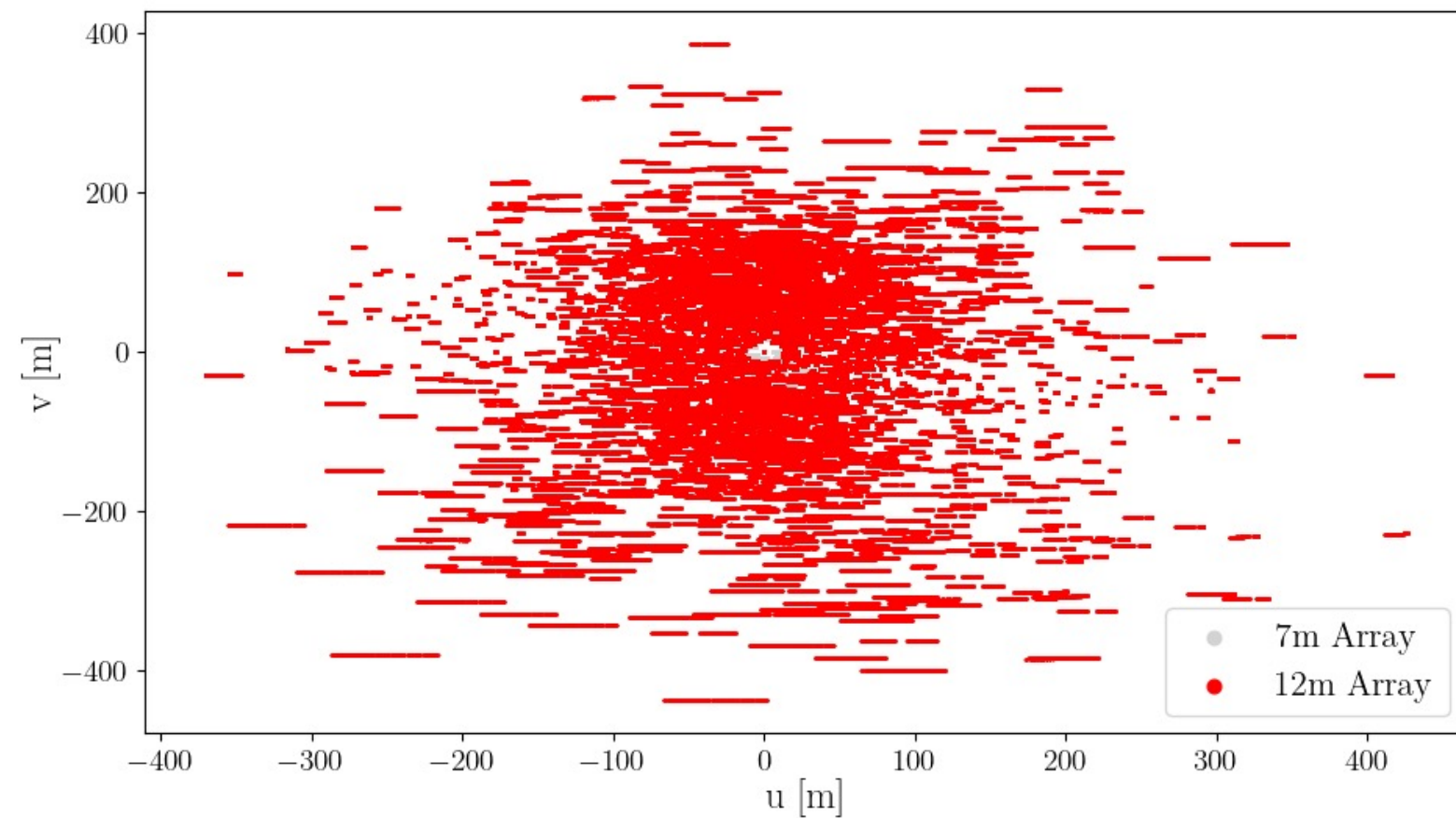


The Atacama Large (Sub-)Millimetre Array (ALMA) is the world's most advanced ground-based interferometer



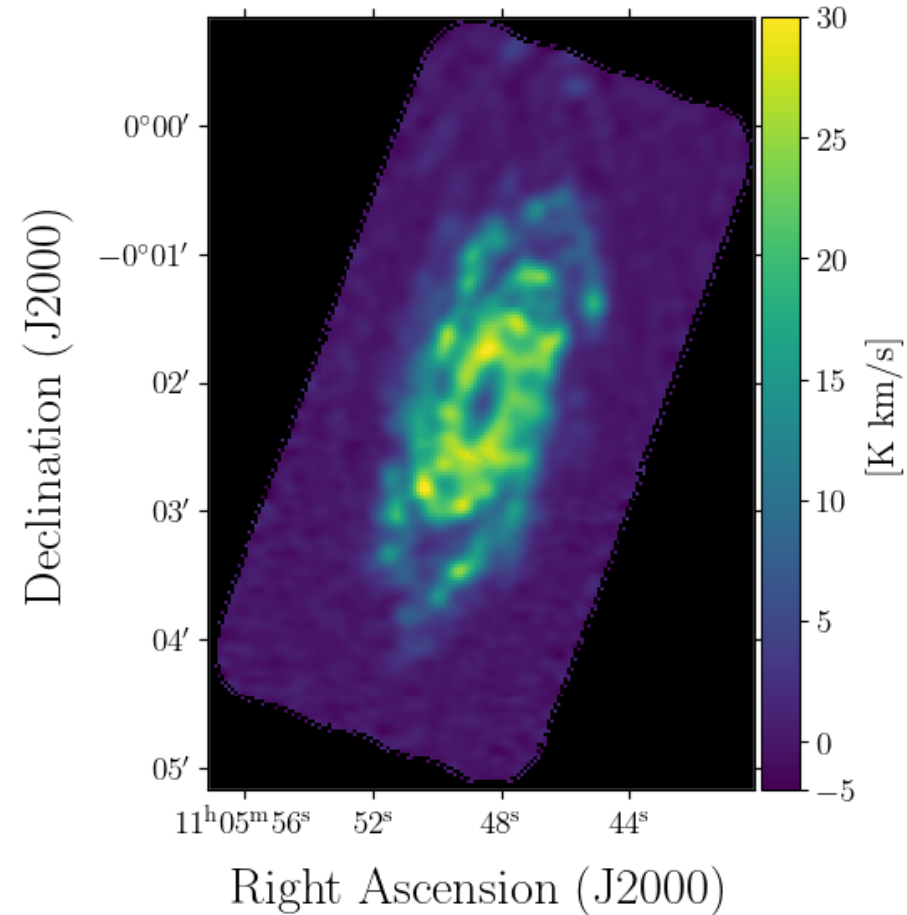
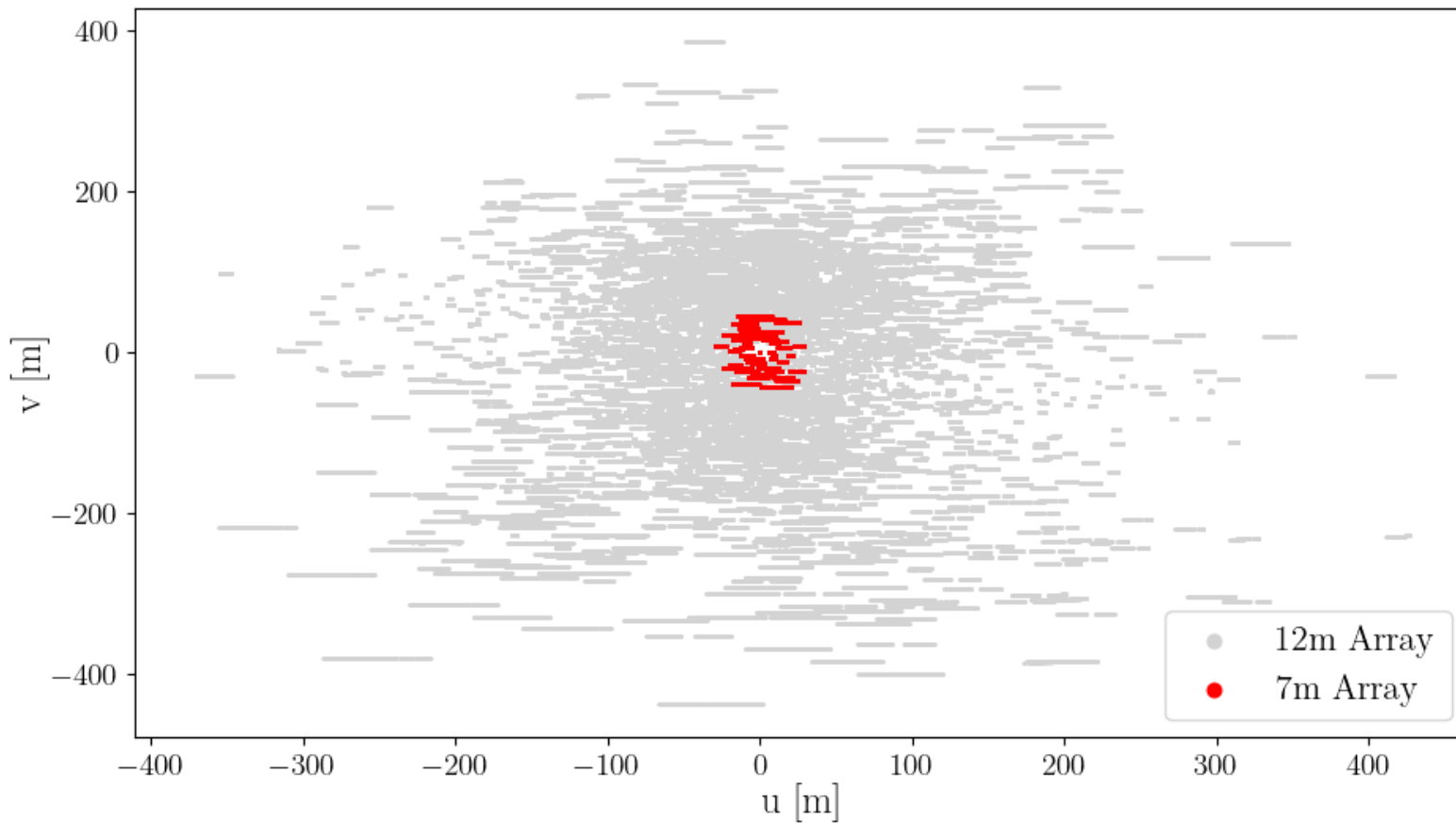
Visibilities are produced by interferometers

Long baselines = high angular resolution



Visibilities are produced by interferometers

Short baselines = low angular resolution



The imaging process has several parameters that introduce biases and artifacts:

- Weighting
 - Robust parameter, uvtaper
- Gridding
- Deconvolver
 - Scales, scale bias
- Masking
- Major/minor cycle length
- Gain

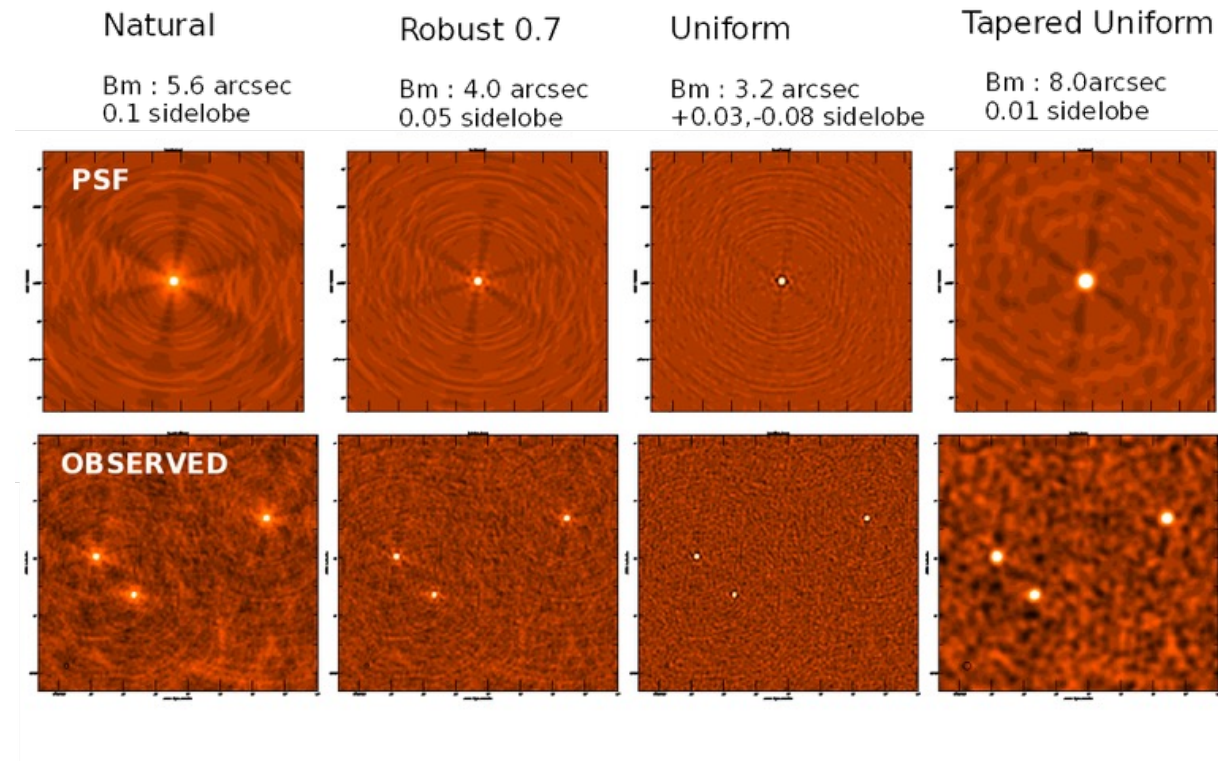
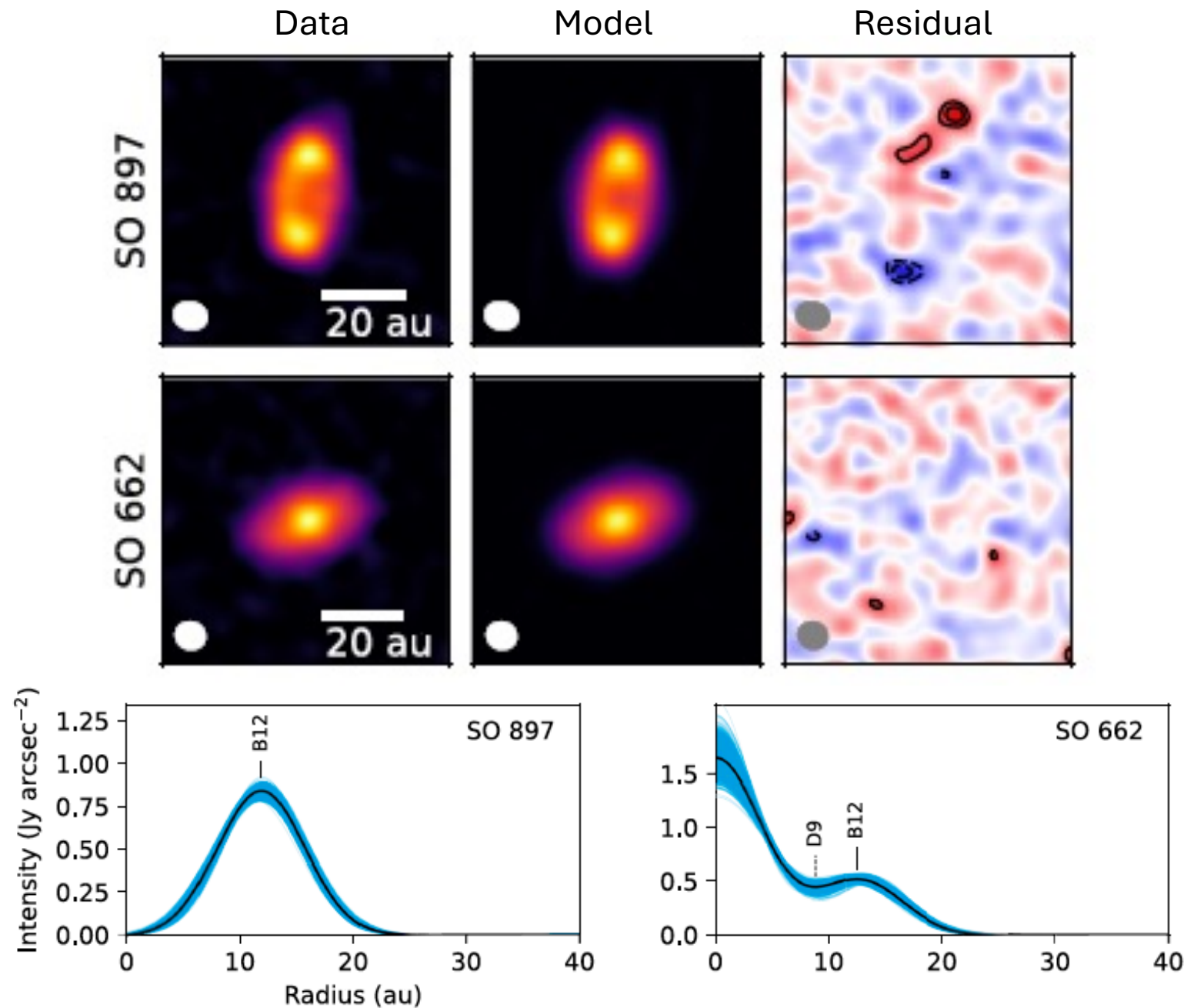


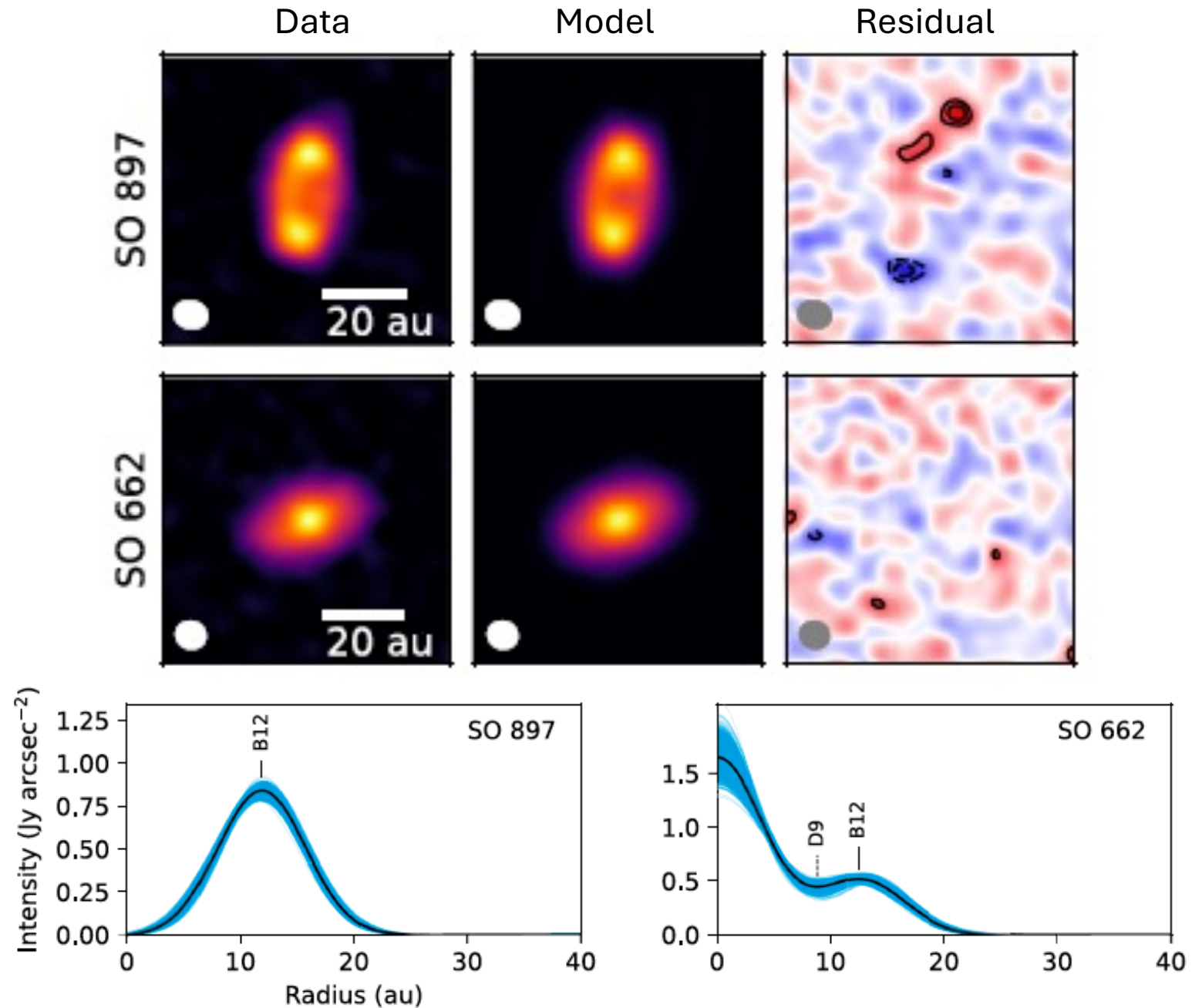
Figure from *Synthesis Imaging*, CASAdocs

Model visibilities
can be fit directly
to observed data



Visibility
modelling can:

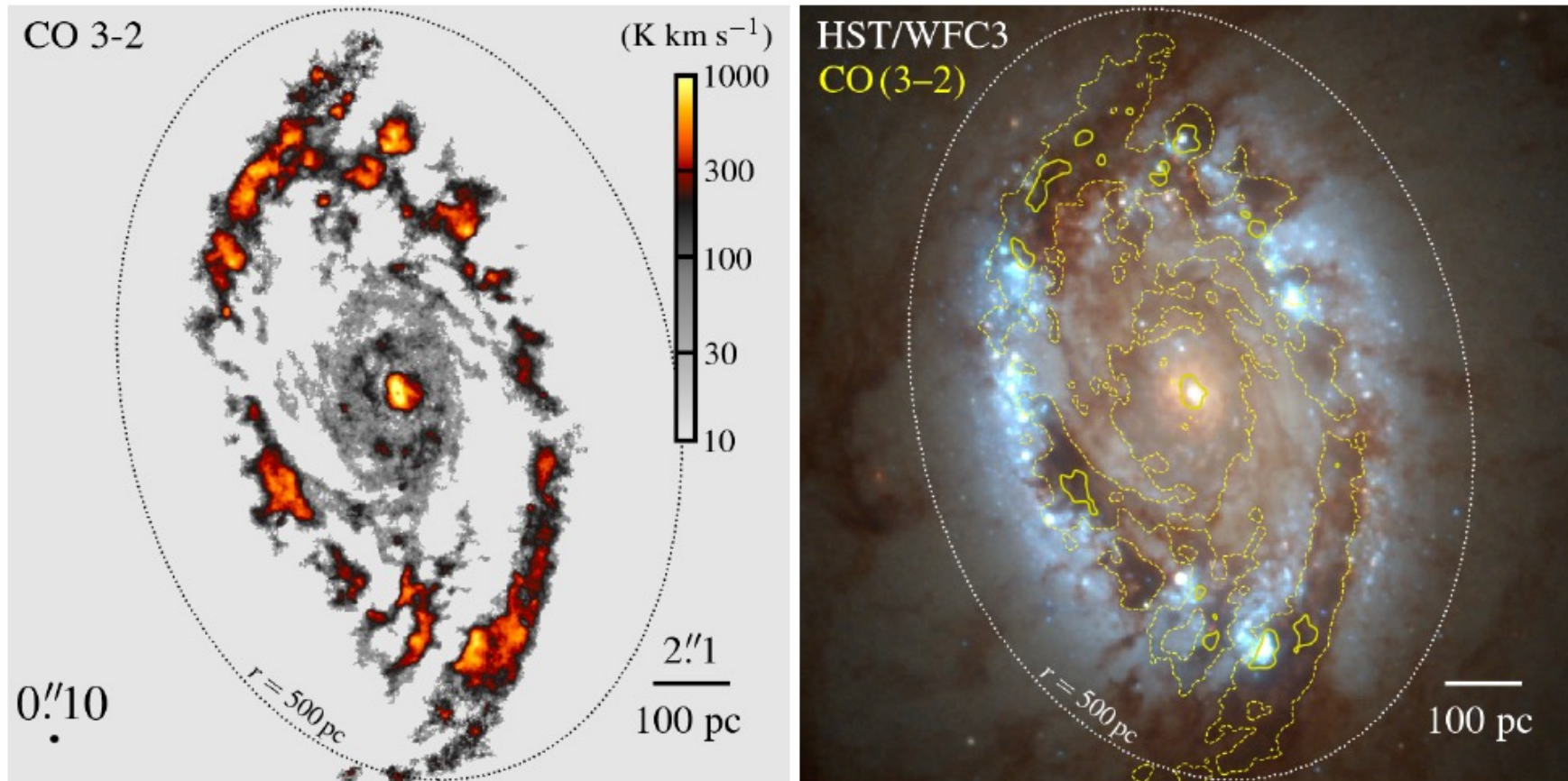
- 1) reveal sub-beam features
- 2) isolate asymmetries in the residual



Star-forming gas ring

Star formation is obscured by dust

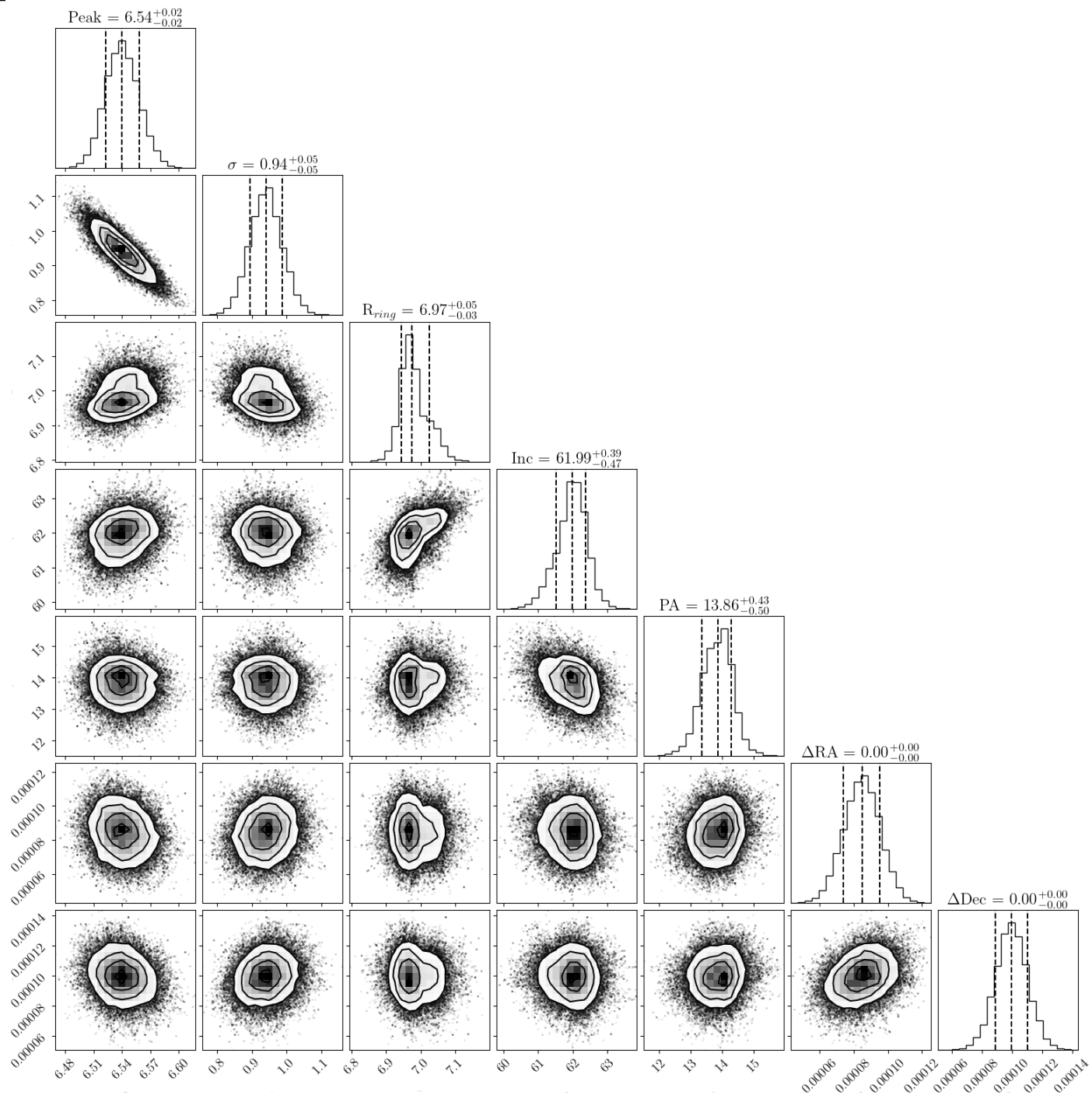
93 GHz star formation not affected by dust



Visibility Fitting Method

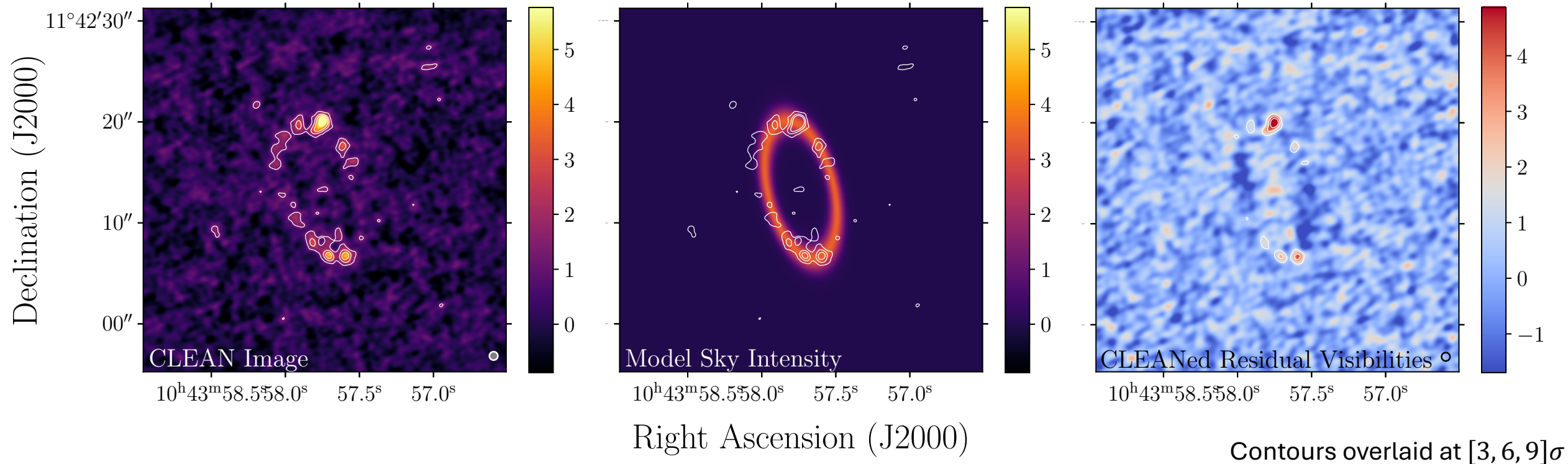
emcee was used to implement an MCMC sampler for model fitting

- 100 walkers
- 150 step burn-in phase
- 3000 step fitting phase



Best Fit Ring Model

93GHz Continuum Intensity of the Star-Forming Ring in NGC 3351 (MJy/sr)

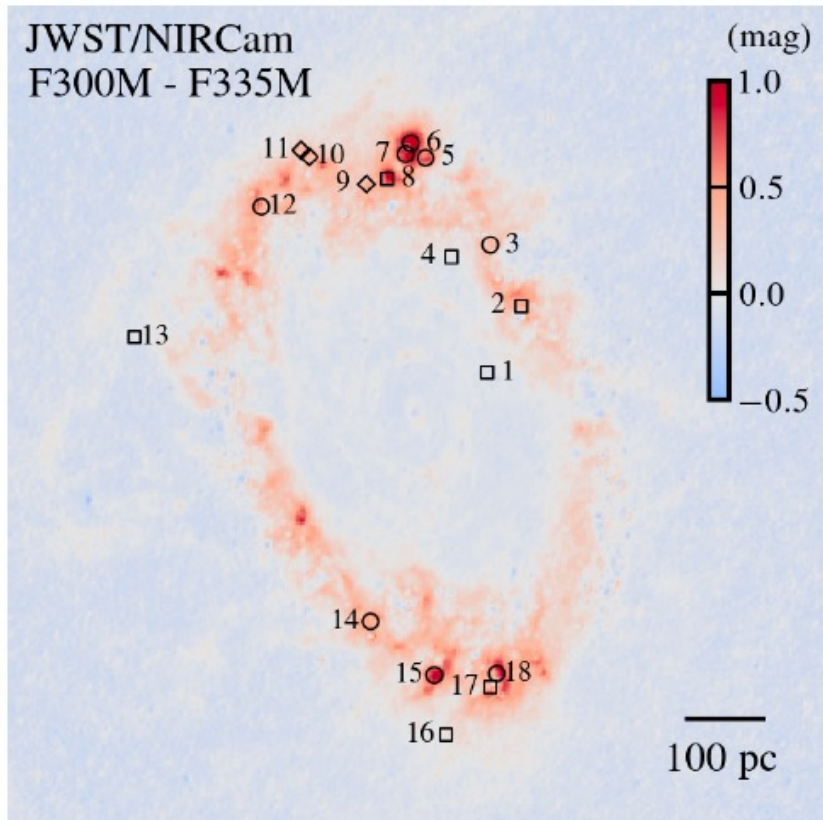


Overall ring structure reproduced in the best fit model

- Significant residuals remain for the point sources along the ring

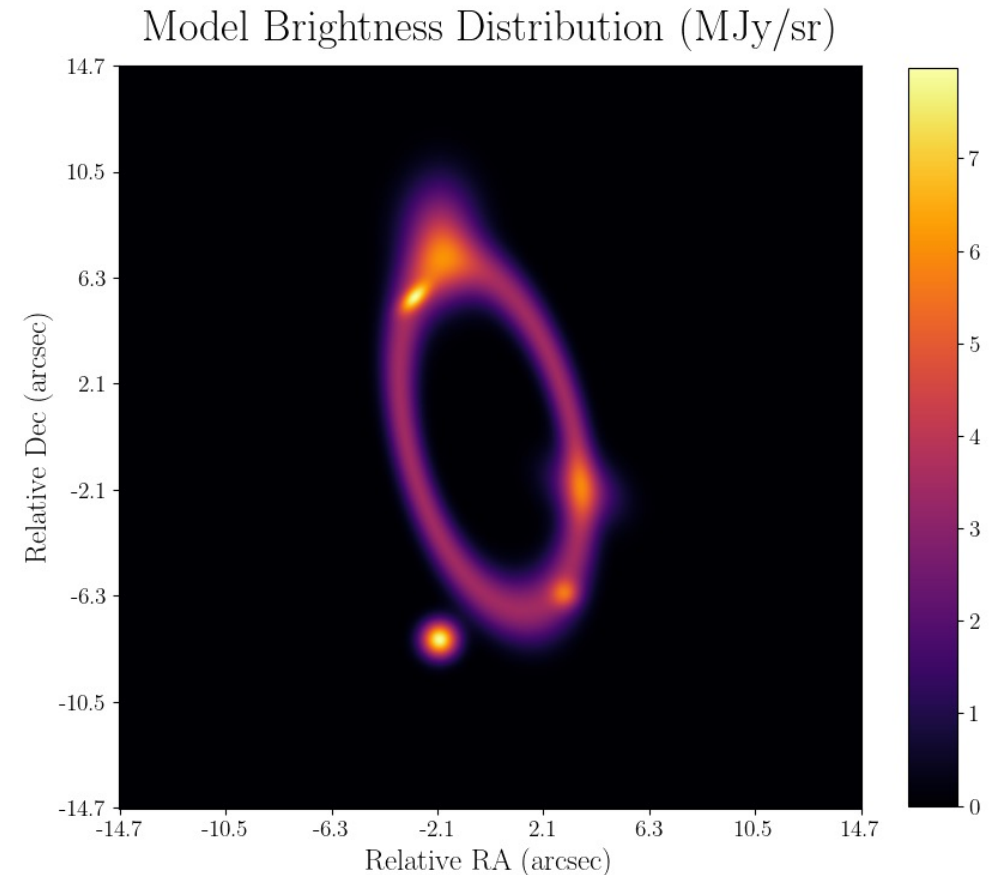
Future Directions

Multi-component models will allow more complex and accurate measurements of galaxy morphology.



Left: NIRCam image of the inner ring in NGC 3351

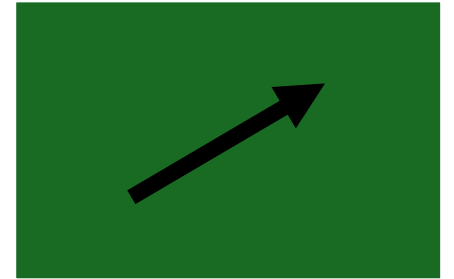
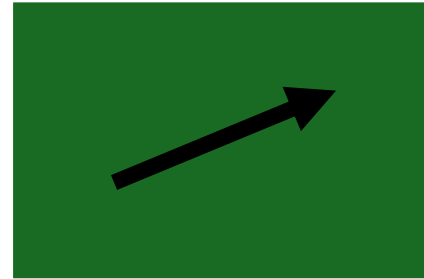
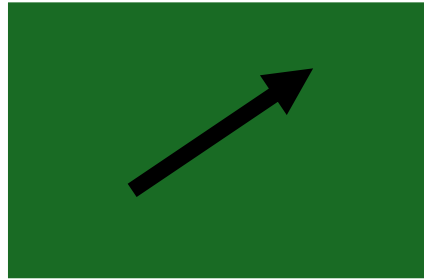
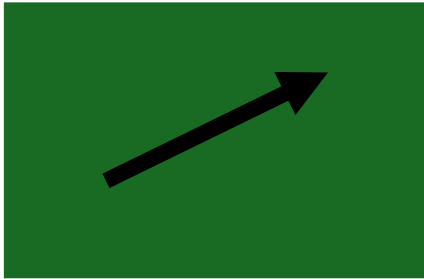
Right: Arbitrary example of a fitting model with a ring and 5 gaussian components



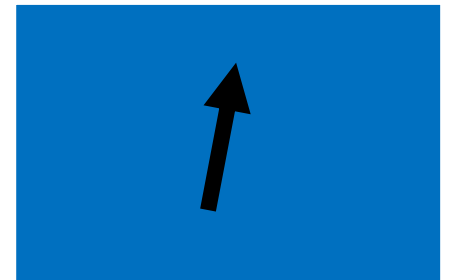
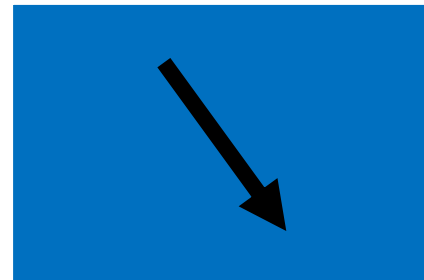
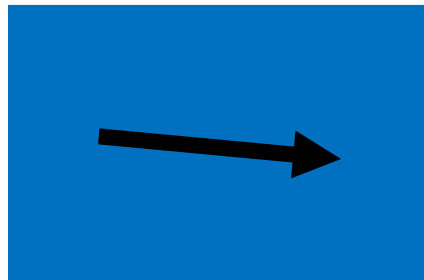
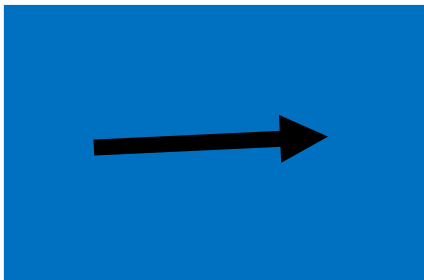
How strongly do the initial parameter guesses affect the fit achieved?
Is this fit the only solution? The best solution?

Converged fits

Model
Equation A

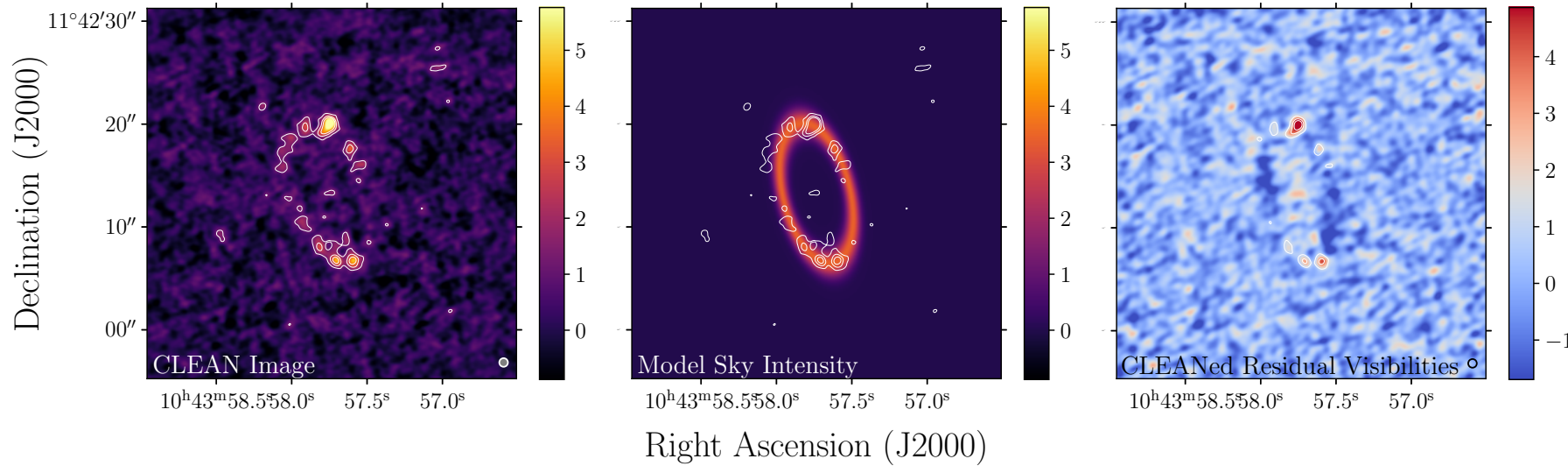


Model
Equation B



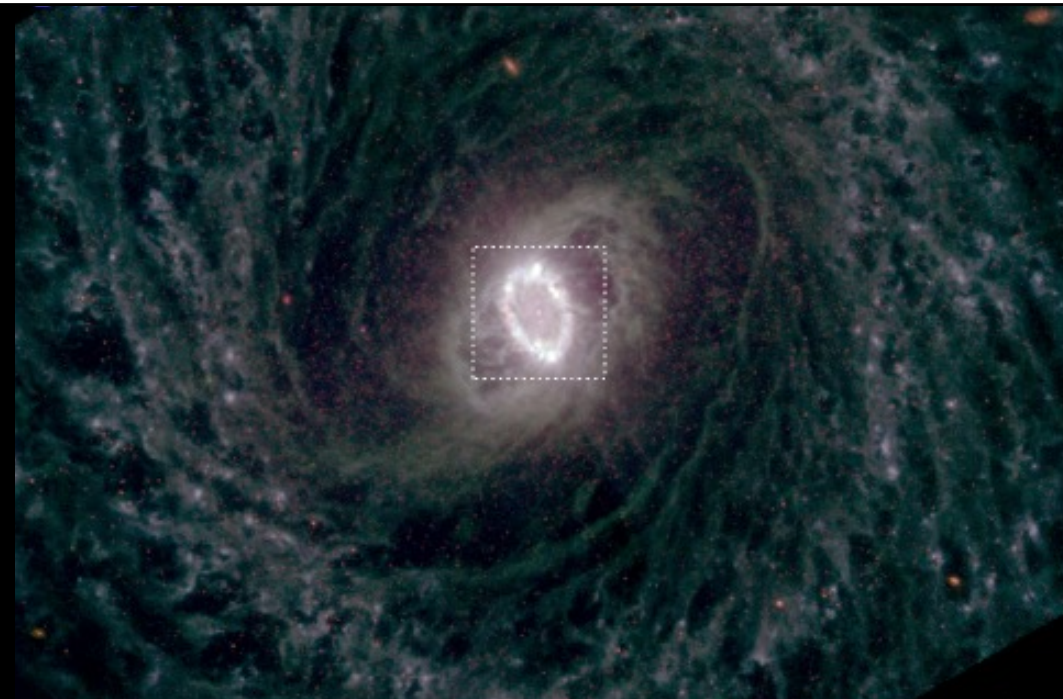
Diverging fits

93GHz Continuum Intensity of the Star-Forming Ring in NGC 3351 (MJy/sr)



Visibility modelling is
an effective technique
for studying the
morphology and
brightness of
star-forming regions
in galaxies

Future work is necessary to make a
more comprehensive tool



Thank you to Doug and Toby for all their help and guidance across the many stages of this project.

Thank you to Helen Kirk, Samuel Fielder and Jess Speedie for their help in understanding imaging with ALMA.